

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Database Management Systems

COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO1: Analyse DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)

Activity Tool : Concept Mapping (Medium)

Assessment Tool Weightage: 20%

QUESTIONS		
Q.No	Question	Bloom's Level
1	Construct a concept map linking the following terms: DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship.	BL2 – Understand
2	Create a concept map for the three-level ANSI/SPARC architecture showing data independence at each level.	BL2 – Understand
3	Map the relationship between file-based systems and database systems, highlighting the problems solved by DBMS.	BL2 – Understand
4	Design a concept map for ER diagram components: entity, weak entity, attribute types, relationship types, and participation constraints.	BL2 – Understand
5	Construct a concept map comparing Hierarchical, Network, and Relational data models with their key properties.	BL2 – Understand

6	Develop a concept map for the EER model showing how it extends the basic ER model with generalization, specialization, and aggregation.	BL2 – Understand
7	Create a visual concept map for the DBMS software architecture, illustrating the flow from user request to data retrieval.	BL2 – Understand
8	Map the data models discussed in CO1 to real-world applications and explain the suitability of each model.	BL2 – Understand
9	Construct a concept map for schemas and instances, showing how they relate to physical and logical data independence.	BL2 – Understand
10	Design a concept map for the roles and responsibilities in a DBMS environment: DBA, End User, Application Developer.	BL1 – Remember

Code of Conduct:

I, Monika R(192324281) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Monika R

CSA0509 – DATABASE
MANAGEMENT SYSTEM

CO1-AT2

CONCEPT MAPPING

NAME: MONIKA R

REG.NO: 192324281

Submitted To:

Kesavan R

Submitted on:

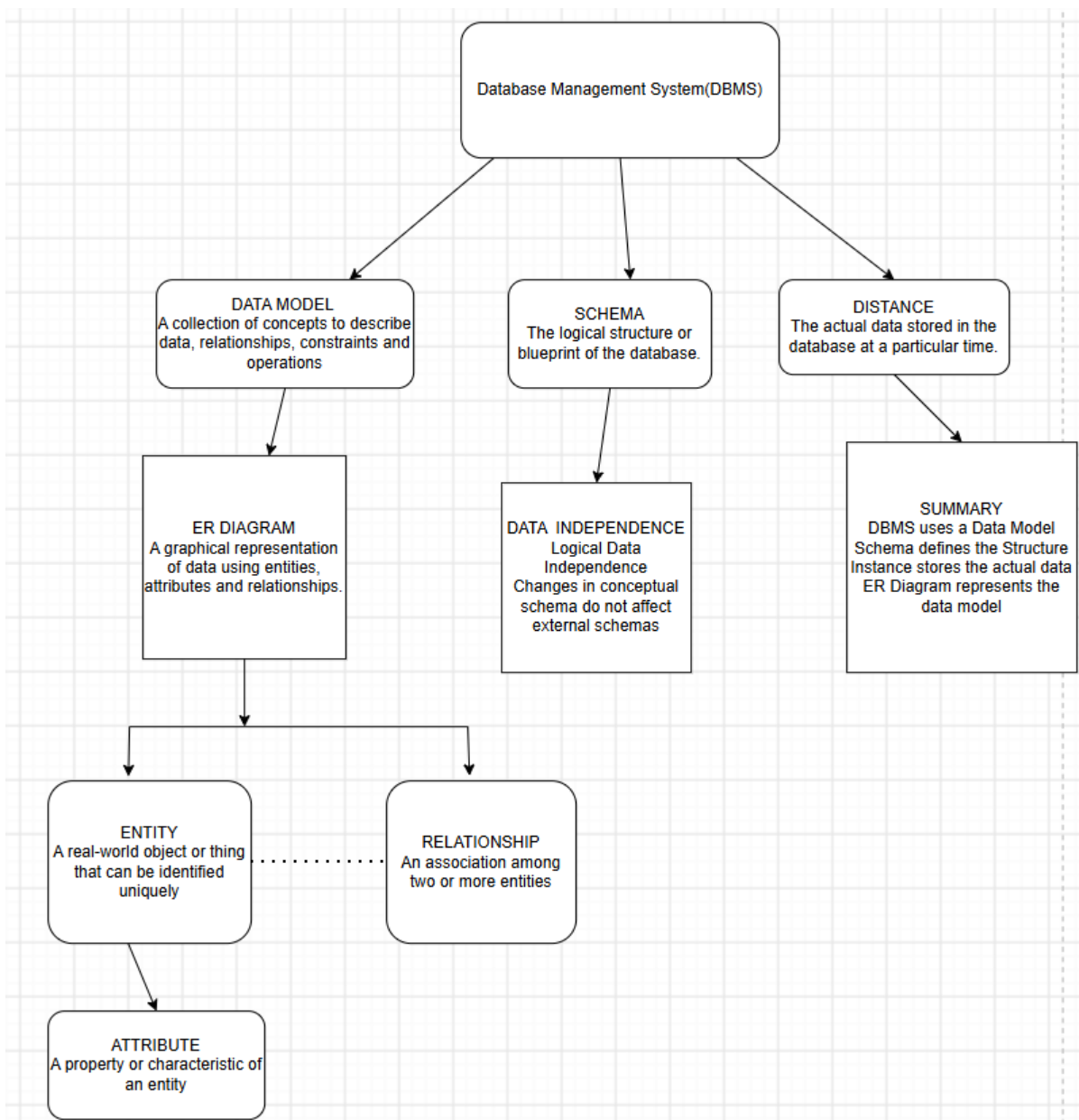
20-07-2026

QUESTION 1:

Construct a concept map linking the following terms: DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship.

ANSWER:

Concept map linking DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship:

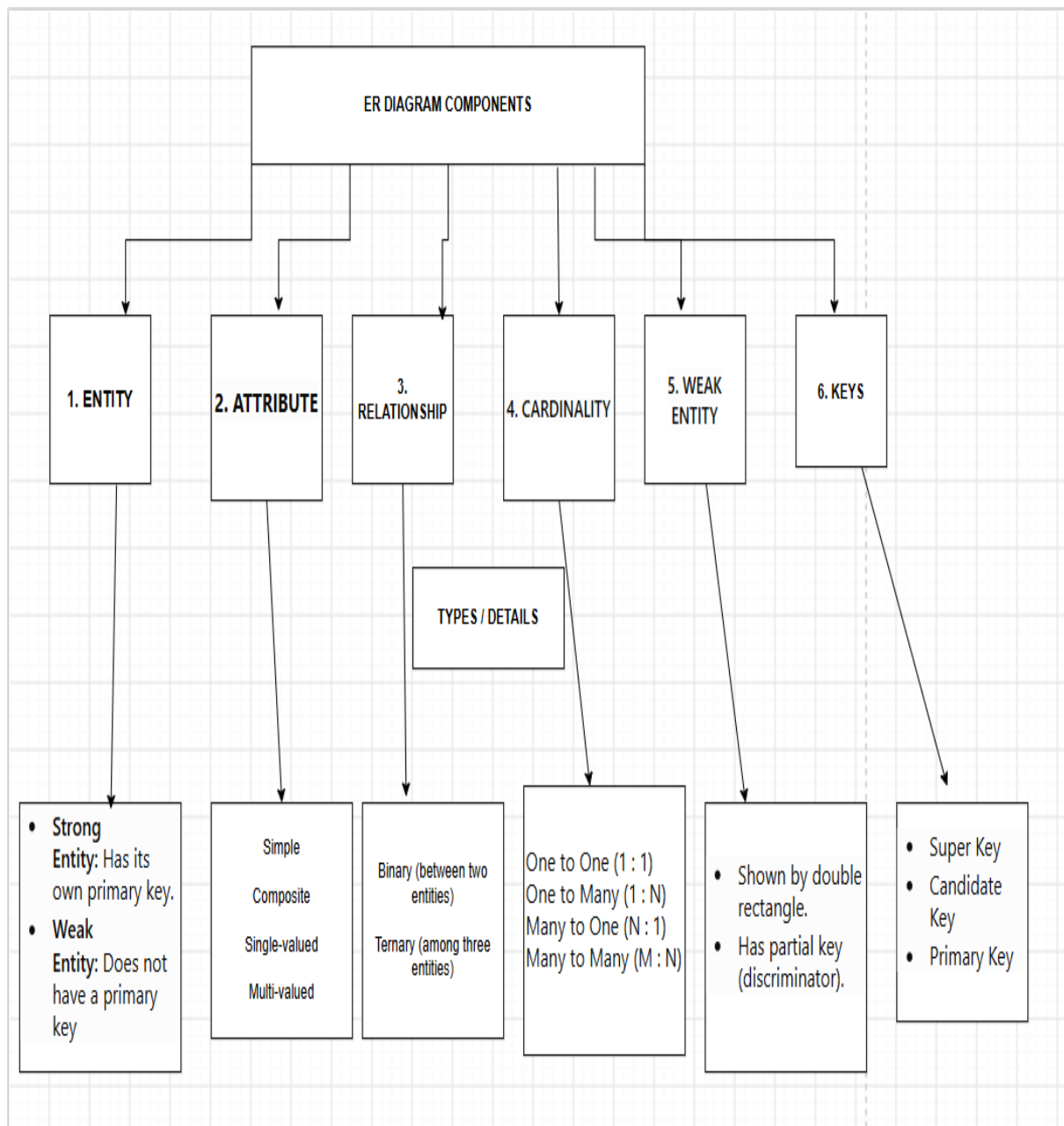


QUESTION:4

Design a concept map for ER diagram components: entity, weak entity, attribute types, relationship types, and participation constraints.

ANSWER:

Concept map for ER diagram components: entity, weak entity, attribute types, relationship types, and participation constraints.

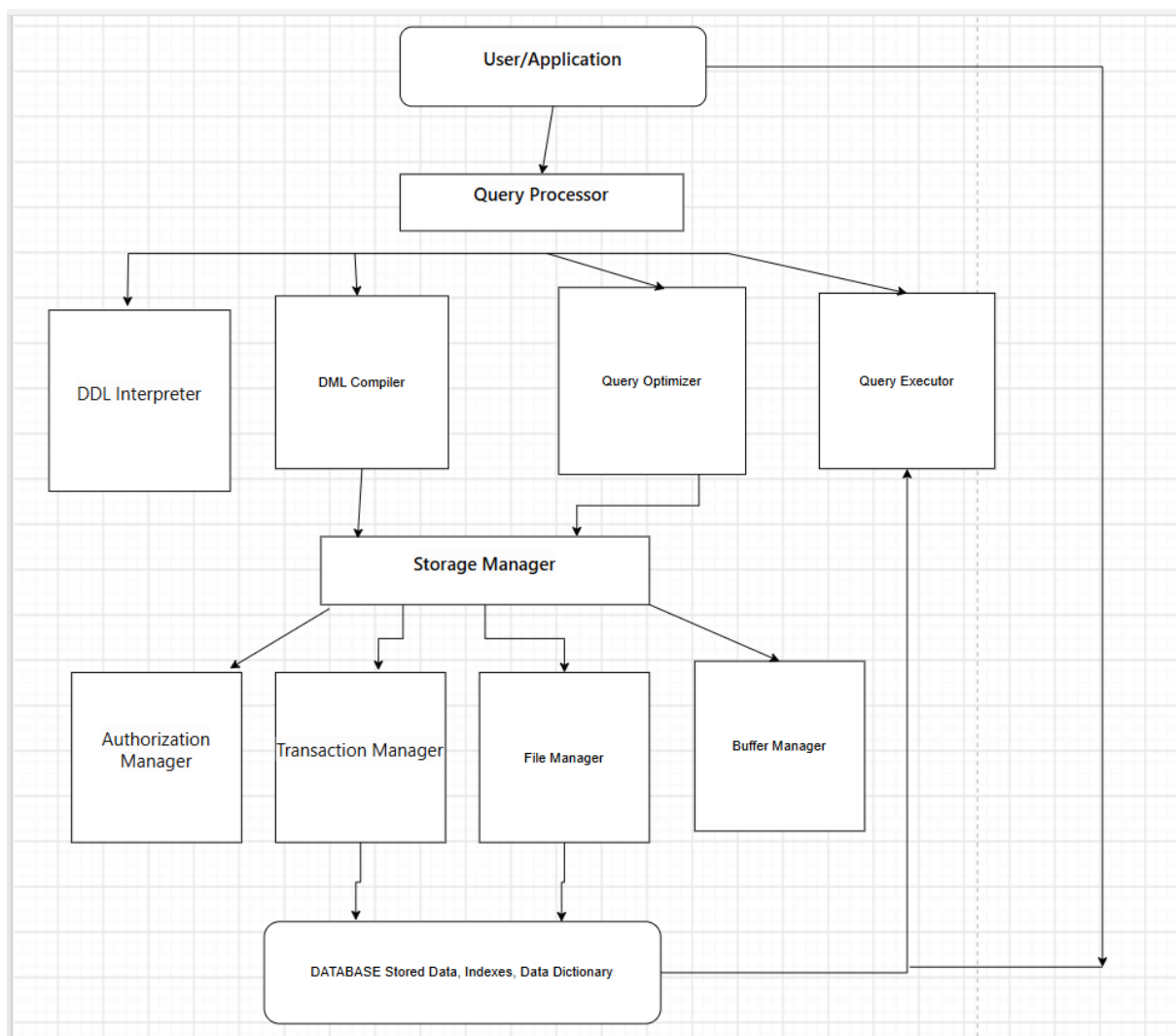


QUESTION:7

Create a visual concept map for the DBMS software architecture, illustrating the flow from user request to data retrieval.

ANSWER:

Visual concept map for the DBMS software architecture:

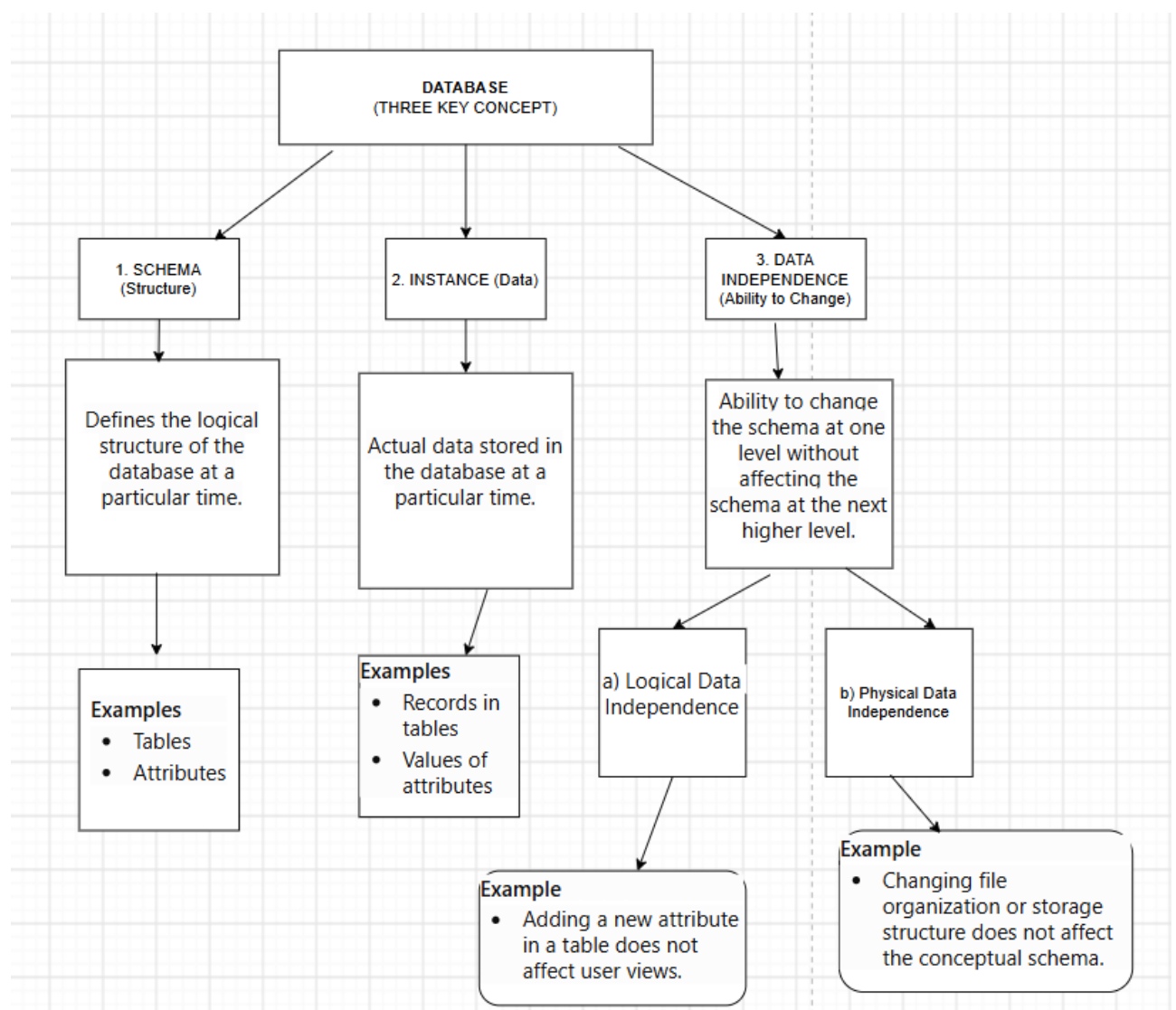


QUESTION:9

Construct a concept map for schemas and instances, showing how they relate to physical and logical data independence.

ANSWER:

Concept map for Schemas and Instances:



QUESTION:10

Design a concept map for the roles and responsibilities in a DBMS environment: DBA, End User, Application Developer.

ANSWER:

Concept map for the roles and responsibilities in a DBMS environment: DBA, End User, Application Developer:

